

From this lab, I learned about diffusion models and how they are used in generating images. Diffusion models involve adding noise to an image and then learning how to reverse this noise in order to get back the image. I was also able to understand the theory and mathematics behind this concept. I was also able to implement a U-Net architecture, which helped in producing better images by retaining significant features in the image.

During my experiments, I found that the training process for diffusion models relies on noise and data. When proper settings are used, better and clearer images are obtained, but this takes more time and requires high computing power. I also used CLIP for image evaluation, which helped in measuring how well the image corresponds to its description. Diffusion models are able to produce better and more diverse images, but it has time consuming process and need computing power.